Notes: Leuz, Nanda, and Wysocki (JFE, 2003)
Earnings Management and Investor Protection: An International Comparison

Contents

1 Big picture

- **Research question.** Why does earnings management differ systematically across countries? More specifically, do firms in countries with stronger investor protection and legal enforcement engage in less earnings management?
- **Main idea.** Insiders use financial reporting discretion to conceal firm performance and protect private control benefits. If outside investors have stronger rights and courts enforce those rights, insiders have fewer private benefits to hide and weaker incentives to manage earnings.
- **Empirical object.** The paper constructs four country-level earnings-management proxies using firm accounting data: two smoothing measures, one accrual-discretion measure, and one loss-avoidance measure.
- **Main finding.** Earnings management is lower in countries with stronger outsider rights, better legal enforcement, larger equity markets, more dispersed ownership, and greater disclosure.
- **Main mechanism.** Investor protection reduces insiders' private control benefits; lower private benefits reduce incentives to hide performance through accounting choices.
- **Important distinction.** The paper does not treat accounting quality as exogenous. It argues that accounting quality is partly an endogenous outcome of the legal and governance environment.
- **Economic takeaway.** Legal institutions affect not only financing and ownership structures, but also the credibility of reported earnings.

2 Incremental contribution of the paper

- **From accounting rules to actual reporting behavior.** Prior international accounting work often compares formal rules. Leuz, Nanda, and Wysocki instead measure realized earnings-management outcomes.
- **Governance-based theory of earnings quality.** The paper links earnings management to the agency conflict between insiders and outside investors. Accounting opacity is interpreted as a tool for protecting private control benefits.
- **Large cross-country sample.** The analysis uses 70,955 firm-year observations, 8,616 nonfinancial firms, and 31 countries over 1990–1999.
- **Multiple earnings-management measures.** The paper avoids relying on one noisy proxy by combining smoothing, accrual magnitude, and small-loss avoidance measures into an aggregate rank score.
- **Institutional clustering.** The paper shows that countries form meaningful clusters: outsider economies with strong markets and enforcement have the lowest earnings management; insider economies with weak enforcement have the highest.

- **Evidence on private control benefits.** The regression evidence links earnings management directly to block-premium measures of private control benefits.
- **Endogeneity insight.** The paper emphasizes that disclosure, earnings quality, ownership concentration, and financial-market development may be jointly determined by investor protection.

3 Economic mechanism

3.1 Insiders, outsiders, and private control benefits

The paper starts from the conflict between insiders and outside investors. Insiders include managers and controlling shareholders. They can extract private control benefits through perks, tunneling, related-party transactions, or other resource diversion. These benefits are valuable precisely because they are not shared with outside investors.

- If outside investors detect diversion, they may discipline insiders through voting, lawsuits, replacement, or other governance mechanisms.
- Insiders therefore benefit from making firm performance harder to interpret.
- Earnings management can hide losses, smooth bad news, and reduce the probability that outsiders intervene.

3.2 Why investor protection should reduce earnings management

The key prediction is:

$$\text{Earnings management} \quad \downarrow \quad \text{as investor protection} \quad \uparrow .$$

The logic has two steps:

- Strong shareholder rights and legal enforcement reduce insiders' ability to extract private control benefits.
- When there are fewer private benefits to protect, insiders have weaker incentives to obscure performance through accounting choices.

This mechanism differs from a pure "rules" story. Strong accounting standards may matter, but the paper argues that standards are effective only when embedded in a legal system that gives outsiders rights and enforces them.

3.3 Potential countervailing force

The paper acknowledges a possible penalty effect. Holding private control benefits fixed, stronger investor protection could make insiders more eager to hide diversion because detection is more costly. Empirically, however, the cross-country evidence suggests that the dominant effect is the reduction in private benefits, so the overall relation between investor protection and earnings management is negative.

4 Earnings-management measures

4.1 Accruals and cash flow from operations

Because direct cash-flow statements are not widely available internationally during the sample period, the paper computes accruals indirectly. The accrual component of earnings is:

$$\text{Accruals}_{it} = (\Delta CA_{it} - \Delta Cash_{it}) - (\Delta CL_{it} - \Delta STD_{it} - \Delta TP_{it}) - Dep_{it},$$

where CA is current assets, $Cash$ is cash and equivalents, CL is current liabilities, STD is short-term debt in current liabilities, TP is taxes payable, and Dep is depreciation and amortization.

Cash flow from operations is then operating income minus accruals.

4.2 EM1: Smoothing reported operating earnings

EM1 is the country median of the firm-level ratio:

$$\frac{\sigma(\text{Operating income})}{\sigma(\text{Cash flow from operations})}.$$

- A lower ratio means operating earnings are smoother relative to underlying operating cash flow.
- Lower EM1 is interpreted as more earnings smoothing.
- In the aggregate ranking, countries with lower EM1 receive higher earnings-management ranks.

4.3 EM2: Accruals as buffers against cash-flow shocks

EM2 is the country-level Spearman correlation between changes in accruals and changes in cash flow from operations:

$$\rho(\Delta Accruals, \Delta CFO).$$

- Accrual accounting naturally creates some negative correlation.
- A more negative correlation suggests that accruals are used more aggressively to smooth cash-flow shocks.
- Countries with more negative EM2 receive higher earnings-management ranks.

4.4 EM3: Magnitude of accruals

EM3 is the country median of:

$$\frac{|\text{Accruals}|}{|\text{Cash flow from operations}|}.$$

- Larger absolute accruals indicate greater reporting discretion.
- This measure captures earnings management that affects the level of reported performance, not only smoothing.

4.5 EM4: Small-loss avoidance

EM4 is the ratio:

$$\frac{\#\text{ small profits}}{\#\text{ small losses}},$$

where small profits are earnings scaled by total assets in $[0, 0.01]$, and small losses are in $[-0.01, 0)$.

- A high ratio indicates that firms are much more likely to report small profits than small losses.
- This is interpreted as evidence that insiders manage earnings to avoid reporting losses.

4.6 Aggregate earnings-management score

For each measure, countries are ranked so that a higher rank means more earnings management. The aggregate score is the average rank across EM1–EM4.

Interpretation: the aggregate measure is not a structural estimate of manipulation. It is a cross-country summary statistic designed to reduce measurement error across noisy indicators.

5 Data and institutional variables

5.1 Accounting data

The accounting data come from Worldscope and cover publicly traded nonfinancial firms.

- **Sample period:** fiscal years 1990–1999.
- **Countries:** 31 countries.
- **Firms:** 8,616 nonfinancial firms.
- **Firm-year observations:** 70,955.

- **Exclusions:** banks and financial firms are excluded. Argentina, Brazil, and Mexico are excluded from the main sample because of hyperinflation. Chile, New Zealand, and Turkey are excluded because there are too few small-loss observations for EM4.
- **Data requirement:** countries need at least 300 firm-year observations for key accounting variables; firms need at least three consecutive years of statements.

5.2 Country-level institutional variables

The paper uses institutional variables drawn primarily from La Porta et al.

- **Outside investor rights:** anti-director rights index, measuring minority shareholder rights.
- **Legal enforcement:** average of judicial efficiency, rule of law, and corruption index.
- **Importance of equity market:** average rank based on minority-held stock market capitalization, listed firms per population, and IPOs per population.
- **Ownership concentration:** median share owned by the three largest shareholders in the ten largest privately owned nonfinancial firms.
- **Disclosure index:** inclusion or omission of 90 items in annual reports.
- **Legal origin and legal tradition:** English, French, German, and Scandinavian origins; common-law or code-law tradition.

5.3 Additional mechanism and robustness variables

- **Private control benefits:** country-level block premiums from Dyck and Zingales, interpreted as a direct proxy for the value of control benefits.
- **Accounting rules:** Hung's accrual-rules index, measuring whether accounting rules accelerate recognition of economic transactions.
- **Ownership concentration controls:** used to ask whether ownership structure itself explains earnings management after investor protection is included.
- **Economic controls:** per capita GDP, inflation, GDP-growth volatility, firm size, capital intensity, and industry composition are considered in robustness checks.

6 Methodology and empirical specifications

6.1 Descriptive country rankings

The paper first compares the four EM measures and the aggregate score across countries. This step establishes the basic cross-country fact: earnings management varies substantially across legal and financial systems.

- High earnings-management countries include Austria, Greece, South Korea, Portugal, and Italy.
- Low earnings-management countries include the United States, Australia, Ireland, Canada, Norway, Sweden, and the United Kingdom.
- The ranking suggests that Anglo-American outsider economies have lower earnings management than many Continental European and Asian insider economies.

6.2 Cluster analysis

The paper conducts a k-means cluster analysis using nine institutional variables. It identifies three country clusters:

- **Cluster 1: outsider economies.** Large stock markets, dispersed ownership, strong outsider rights, strong enforcement, and extensive disclosure.
- **Cluster 2: insider economies with stronger enforcement.** Concentrated ownership and smaller stock markets, but relatively strong legal enforcement.
- **Cluster 3: insider economies with weaker enforcement.** Concentrated ownership, weaker enforcement, smaller stock markets, and the highest earnings management.

The cluster evidence is descriptive, but it makes the institutional pattern transparent.

6.3 Cross-country rank regressions

The main cross-country regression relates the aggregate earnings-management rank to investor-protection variables:

$$EM_c = \alpha + \beta_1 \text{OutsideRights}_c + \beta_2 \text{LegalEnforcement}_c + \varepsilon_c.$$

The hypothesis is:

$$\beta_1 < 0, \qquad \beta_2 < 0.$$

That is, stronger minority shareholder rights and stronger legal enforcement should be associated with lower earnings management.

6.4 Instrumental variables

To address endogeneity of investor protection, the paper estimates 2SLS regressions using legal origin and pre-sample country wealth as instruments.

- Legal origin is treated as historically predetermined.
- Pre-sample GDP helps capture the resources needed to build and maintain a legal infrastructure.

- The 2SLS results continue to show a negative relation between investor protection and earnings management.

6.5 Private control benefits regression

The paper also estimates the relation between earnings management and private control benefits:

$$EM_c = \alpha + \theta \text{PrivateControlBenefits}_c + \varepsilon_c.$$

The predicted sign is positive. If insiders have more valuable private benefits to protect, they have stronger incentives to obscure firm performance.

6.6 Robustness checks

The paper examines whether the main results survive controls for:

- per capita GDP;
- firm size and industry composition;
- inflation and macroeconomic growth volatility;
- accounting rules and tax-book conformity;
- ownership concentration;
- exclusion of individual countries, including the United States.

The main investor-protection result remains robust.

7 Identification and interpretation

7.1 What the paper can identify

The evidence identifies a strong cross-country association between legal investor protection and realized earnings-management outcomes. The paper's causal interpretation is supported by instrumental variables and robustness checks, but the design remains observational and country-level.

7.2 Why multiple EM measures matter

Earnings management is hard to observe directly. The paper therefore uses several imperfect proxies. The fact that the four measures are highly correlated and produce similar country rankings strengthens the interpretation that they capture a common underlying reporting-quality dimension.

7.3 Why investor protection is interpreted as primitive

The paper views investor protection as more fundamental than disclosure or ownership concentration because legal rights and enforcement shape insiders' ability to extract control benefits. Under this view, earnings quality, disclosure, ownership dispersion, and equity-market development are complementary outcomes of the institutional environment.

7.4 Endogeneity concerns

Important limitations remain:

- legal institutions may be correlated with unobserved culture, politics, or economic development;
- accounting rules and enforcement may be jointly determined with reporting behavior;
- country-level regressions have small sample sizes;
- Worldscope coverage varies across countries and is tilted toward listed firms.

The IV strategy helps but does not eliminate all concerns.

7.5 Interpreting the main coefficient

The negative coefficient on outside investor rights should be read as evidence that countries with stronger shareholder protection have lower country-level earnings-management ranks. It is not a firm-level structural estimate of how any single legal rule changes accrual manipulation.

8 Tables and figures

8.1 Table 1: Descriptive statistics of sample firms and countries

Read:

- Table 1 documents the 31-country sample, the number of firm-year observations, firm size, capital intensity, manufacturing share, GDP per capita, inflation, and GDP-growth volatility.
- Japan has the largest number of firm-year observations, while the United States sample is limited to large firms in Worldscope.
- There is substantial cross-country variation in firm size, macroeconomic conditions, and industry composition, motivating later controls.

Economic message: The paper studies reporting behavior across very different country environments. The variation in economic development and firm characteristics makes controls important.

8.2 Table 2: Earnings-management measures and institutional characteristics

Read Panel A:

- Panel A ranks countries using four earnings-management proxies.
- Austria and Greece have the highest aggregate scores, while the United States has the lowest score.
- Continental European and Asian countries generally show more smoothing and loss avoidance than Anglo-American countries.

Read Panel B:

- Panel B lists legal origin, legal tradition, outside investor rights, legal enforcement, equity-market importance, ownership concentration, and disclosure index.
- Countries with low earnings-management scores tend to have stronger outsider rights, stronger enforcement, and larger equity markets.

Read Panel C:

- Aggregate earnings management is negatively correlated with outside investor rights, equity-market importance, and disclosure.
- It is positively correlated with ownership concentration.
- The simple correlation with legal enforcement is negative but less statistically strong.

Economic message: The basic cross-country patterns line up with the governance mechanism: earnings management is higher when ownership is concentrated and investor protection is weak.

8.3 Table 3: Earnings management and institutional clusters

Read:

- The cluster analysis separates countries into three institutional groups.
- Cluster 1 has outsider features: large stock markets, low ownership concentration, high disclosure, strong enforcement, and low earnings management.
- Cluster 3 has insider features and weak enforcement; it has the highest average earnings-management score.

- The mean aggregate EM score is 10.1 in Cluster 1, 16.1 in Cluster 2, and 20.6 in Cluster 3.

Economic message: Earnings management is not randomly distributed across countries. It varies systematically with broad governance institutions.

8.4 Tables 4 and 5: Investor protection, private control benefits, and robustness

Read Table 4:

- Outside investor rights and legal enforcement are negatively associated with aggregate earnings management.
- The relation remains in 2SLS specifications using legal origin and pre-sample GDP as instruments.
- Private control benefits are positively associated with earnings management.

Read Table 5:

- The relation between investor protection and earnings management remains after controlling for accounting rules.
- Accrual rules matter in a simple rank regression but lose significance in the 2SLS specification.
- Ownership concentration does not have incremental explanatory power once investor protection is included.

Economic message: The evidence supports the interpretation that investor protection is a deeper determinant of earnings quality than formal accounting rules alone or average ownership concentration.

Table 1

Descriptive statistics of sample firms and countries

The full sample consists of 70,955 firm-year observations for the fiscal years 1990 to 1999 across 31 countries and 8,616 non-financial firms. Financial accounting information is obtained from the November 2000 version of the Worldscope Database. To be included in our sample, countries must have at least 300 firm-year observations for a number of accounting variables, including total assets, sales, net income, and operating income. For each firm, we require income statement and balance sheet information for at least three consecutive years. We discard three countries (Chile, New Zealand, Turkey) because of an insufficient number of observations to compute the loss avoidance measure, and three countries (Argentina, Brazil, Mexico) due to hyperinflation. Firm size is measured as total US$ sales (in thousands). Capital intensity is measured as the ratio of long-term assets over total assets. The fraction of manufacturing firms is the percentage of firm-year observations with SIC 2000 to 3999. Average per capita GDP in constant 1995 US$ is computed from 1990 to 1999. Inflation is measured as the average percentage change in consumer prices from 1990 to 1998. Volatility of GDP growth is measured as the standard deviation of the growth rate in real per capita GDP from 1990 to 1998.

Country	# Firm-years	Median firm size in US$	Median capital intensity	Fraction of mfg. firms	Per-capita GDP in US$	Inflation (%)	Volatility of GDP growth (%)
AUSTRALIA	1,483	233,344	0.425	0.319	20,642	2.62	2.01
AUSTRIA	564	213,101	0.313	0.710	29,287	2.62	1.22
BELGIUM	727	277,510	0.280	0.563	27,357	2.26	1.45
CANADA	3,322	271,287	0.465	0.381	19,687	2.25	1.92
DENMARK	1,235	119,113	0.344	0.573	34,163	2.07	1.23
FINLAND	854	308,974	0.345	0.618	26,296	2.25	4.69
FRANCE	4,404	178,163	0.187	0.548	26,960	2.04	1.42
GERMANY	4,440	336,894	0.282	0.637	30,166	2.51	1.46
GREECE	858	38,305	0.295	0.568	11,393	12.06	1.48
HONG KONG	1,483	167,754	0.376	0.513	21,610	4.10	3.89
INDIA	2,064	63,027	0.409	0.859	374	10.09	2.32
INDONESIA	787	75,502	0.361	0.694	961	13.86	7.26
IRELAND	436	124,021	0.386	0.438	18,707	2.38	3.03
ITALY	1,213	350,380	0.280	0.721	19,025	4.40	1.25
JAPAN	16,475	463,191	0.289	0.583	41,200	1.38	2.29
KOREA (SOUTH)	1,692	452,349	0.382	0.724	10,250	6.28	4.64
MALAYSIA	2,036	81,407	0.403	0.557	4,043	3.97	4.35
NETHERLANDS	1,561	349,909	0.333	0.503	27,037	2.48	1.07
NORWAY	988	104,483	0.356	0.410	33,189	2.46	1.28
PAKISTAN	508	24,907	0.432	0.913	488	10.34	2.25
PHILIPPINES	429	60,814	0.460	0.500	1,093	9.80	2.42
PORTUGAL	460	97,229	0.412	0.545	10,942	6.40	1.68
SINGAPORE	1,100	104,187	0.377	0.472	22,721	2.15	2.66
SOUTH AFRICA	1,043	380,644	0.327	0.445	3,914	10.41	1.92
SPAIN	1,082	333,207	0.424	0.492	15,092	4.43	1.64
SWEDEN	1,384	261,343	0.295	0.505	27,350	3.59	2.29
SWITZERLAND	1,320	377,488	0.394	0.626	44,485	2.51	1.65
TAIWAN	1,001	208,798	0.357	0.809	11,893	3.37	0.80
THAILAND	1,529	55,344	0.433	0.578	2,570	5.50	3.28
UNITED KINGDOM	10,685	109,337	0.335	0.430	19,126	3.95	2.03
UNITED STATES	3,792	3,597,429	0.333	0.556	27,836	3.09	1.64
Mean	2,289	316,756	0.358	0.574	19,028	4.76	2.34
Median	1,235	208,798	0.357	0.557	19,687	3.37	1.92
Min	429	24,907	0.187	0.319	374	1.38	0.80
Max	16,475	3,597,429	0.465	0.913	44,485	13.86	7.26

Table 1: Descriptive statistics of sample firms and countries.

Table 2

The variables are computed from 70,955 firm-year observations for fiscal years 1990 to 1999 across 31 countries and 8,616 non-financial firms. Data are obtained from the Worldscope Database (November 2000). EM1 is the country's median ratio of the firm-level standard deviations of operating income and operating cash flow (both scaled by *lagged* total assets). The *cash* flow from operations is equal to operating income minus accruals, where accruals are calculated as: (Δtotal current assets – Δcash)–(Δtotal current liabilities – Δshort-term debt –Δtaxes payable) – depreciation expense. EM2 is the country's Spearman correlation between the change in accruals and the change in cash flow from operations (both scaled by lagged total assets). EM3 is the country's median ratio of the absolute value of accruals and the absolute value of the cash flow from operations. EM4 is the number of "small profits" divided by the number of "small losses" for each country. A firm-year observation is classified as a small profit if net earnings (scaled by lagged total assets) are in the range [0,0.01]. A firm-year observation is classified as a small loss if net earnings (scaled by lagged total assets) are in the range [−0.01,0). Net earnings are bottom-line reported income after interest, taxes, special items, extraordinary items, reserves, and any other item. The aggregate earnings management score is the average rank across all four measures, EM1-EM4. The sign in the column heading indicates whether *higher* scores for the respective EM measure imply *more* earnings management (+) or *less* earnings management (–).

Panel A: Country scores for earnings management measures (Sorted by aggregate earnings management)

	Earnings smoothing measures		Earnings discretion measures		
	EM1σ(OpInc)/σ(CFO) (–)	EM2 ρ(ΔAcc, ΔCFO) (–)	EM3 |Acc|/|CFO| (+)	EM4 # of SmProfit/# of SmLoss (+)	Aggregate earnings management score
AUSTRIA	0.345	−0.921	0.783	3.563	28.3
GREECE	0.415	−0.928	0.721	4.077	28.3
KOREA (SOUTH)	0.399	−0.922	0.685	3.295	26.8
PORTUGAL	0.402	−0.911	0.745	3.000	25.1
ITALY	0.488	−0.912	0.630	4.154	24.8
TAIWAN	0.431	−0.898	0.646	2.765	22.5
SWITZERLAND	0.473	−0.873	0.547	5.591	22.0
SINGAPORE	0.455	−0.882	0.627	3.000	21.6
GERMANY	0.510	−0.867	0.848	3.006	21.5
JAPAN	0.560	−0.905	0.567	3.996	20.5
BELGIUM	0.526	−0.831	0.677	3.571	19.5
HONG KONG	0.451	−0.850	0.552	3.545	19.5

Table 2, Panel A, part 1: Country scores for earnings management.

INDIA	0.523	−0.867	0.509	6.000	19.1
SPAIN	0.539	−0.865	0.514	6.000	18.6
INDONESIA	0.481	−0.825	0.506	7.200	18.3
THAILAND	0.602	−0.868	0.671	3.136	18.3
PAKISTAN	0.508	−0.913	0.513	2.643	17.8
NETHERLANDS	0.491	−0.861	0.480	3.313	16.5
DENMARK	0.559	−0.875	0.526	2.708	16.0
MALAYSIA	0.569	−0.857	0.578	2.658	14.8
FRANCE	0.561	−0.845	0.579	2.370	13.5
FINLAND	0.555	−0.818	0.517	2.633	12.0
PHILIPPINES	0.722	−0.804	0.555	2.455	8.8
UNITED KINGDOM	0.574	−0.807	0.397	1.802	7.0
SWEDEN	0.621	−0.764	0.466	2.568	6.8
NORWAY	0.713	−0.722	0.556	1.235	5.8
SOUTH AFRICA	0.643	−0.840	0.297	1.667	5.6
CANADA	0.649	−0.759	0.478	2.338	5.3
IRELAND	0.607	−0.788	0.371	1.667	5.1
AUSTRALIA	0.625	−0.790	0.450	1.486	4.8
UNITED STATES	0.765	−0.740	0.311	1.631	2.0
Mean	0.541	−0.849	0.558	3.196	
Median	0.539	−0.861	0.552	3.000	
Standard Deviation	0.100	0.056	0.128	1.413	
Min	0.345	−0.928	0.297	1.235	
Max	0.765	−0.722	0.848	7.200	

Table 2, Panel A, part 2: Continued country scores.

Panel B: Institutional characteristics of the sample countries (Sorted by aggregate earnings management)

Countries are sorted based on the aggregate earnings management score tabulated in Panel A of Table 2. The classification of the Legal Origin and the Legal Tradition are based on La Porta et al., (1998). CD (CM) indicates a code-law (common-law) country. The Outside Investor Rights variable is the anti-director rights index created by La Porta et al. (1998); it is an aggregate measure of minority shareholder rights and ranges from zero to five. Legal Enforcement is measured as the mean score across three legal variables used in La Porta et al. (1998): (1) the efficiency of the judicial system, (2) an assessment of rule of law, and (3) the corruption index. All three variables range from zero to ten. The Importance of Equity Market is measured by the mean rank across three variables used in La Porta et al. (1997): (1) the ratio of the aggregate stock market capitalization held by minorities to gross national product, (2) the number of listed domestic firms relative to the population, and (3) the number of IPOs relative to the population. Each variable is ranked such that higher scores indicate a greater importance of the stock market. Ownership Concentration is measured as the median percentage of common shares owned by the largest three shareholders in the ten largest privately owned non-financial firms (La Porta et al., 1998). The Disclosure Index measures the inclusion or omission of 90 items in the 1990 annual reports (La Porta et al., 1998); it is not available (NA) for three countries in our sample.

Country	Legal Origin	Legal Tradition	Outside Investor Rights	Legal Enforcement	Important of Equity Market	Ownership Concentration	Disclosure Index
AUSTRIA	German	CD	2	9.4	7.0	0.51	54
GREECE	French	CD	2	6.8	11.5	0.68	55
KOREA (SOUTH)	German	CD	2	5.6	11.7	0.20	62
PORTUGAL	French	CD	3	7.2	11.8	0.59	36
ITALY	French	CD	1	7.1	6.5	0.60	62
TAIWAN	German	CD	3	7.4	13.3	0.14	65
SWITZERLAND	German	CD	2	10.0	24.8	0.48	68
SINGAPORE	English	CM	4	8.9	28.8	0.53	78
GERMANY	German	CD	1	9.1	5.0	0.50	62
JAPAN	German	CD	4	9.2	16.8	0.13	65
BELGIUM	French	CD	0	9.4	11.3	0.62	61
HONG KONG	English	CM	5	8.9	28.8	0.54	69
INDIA	English	CM	5	5.6	14.0	0.43	57
SPAIN	French	CD	4	7.1	7.2	0.50	64
INDONESIA	French	CD	2	2.9	4.7	0.62	NA
THAILAND	English	CM	2	4.9	14.3	0.48	64
PAKISTAN	English	CM	5	3.7	7.5	0.41	NA
NETHERLANDS	French	CD	2	10.0	19.3	0.31	64
DENMARK	Scandinavian	CD	2	10.0	20.0	0.40	62
MALAYSIA	English	CM	4	7.7	25.3	0.52	76
FRANCE	French	CD	3	8.7	9.3	0.24	69
FINLAND	Scandinavian	CD	3	10.0	13.7	0.34	77
PHILIPPINES	French	CD	3	3.5	5.7	0.51	65
UNITED KINGDOM	English	CM	5	9.2	25.0	0.15	78

Table 2, Panel B, part 1: Institutional characteristics.

SWEDEN	Scandinavian	CD	3	10.0	16.7	0.28	83
NORWAY	Scandinavian	CD	4	10.0	20.3	0.31	74
SOUTH AFRICA	English	CM	5	6.4	16.3	0.52	70
CANADA	English	CM	5	9.8	23.3	0.24	74
IRELAND	English	CM	4	8.4	17.3	0.36	NA
AUSTRALIA	English	CM	4	9.5	24.0	0.28	75
UNITED STATES	English	CM	5	9.5	23.3	0.12	71

Table 2, Panel B, part 2: Continued institutional characteristics.

Panel C: Correlation between earnings management and institutional characteristics

The table presents Spearman correlations and significance levels (in parentheses) between the following measures. The aggregate earnings management score is the average rank of all four earnings management measures, EM1–EM4. Outside Investor Rights is the anti-director rights index from La Porta et al. (1998). It is an aggregate measure of (minority) shareholder rights and ranges from zero to six. Legal Enforcement is measured as the mean score across three legal variables used in La Porta et al. (1998): (1) the efficiency of the judicial system, (2) an assessment of rule of law, and (3) the corruption index. All three variables range from zero to ten. The Importance of the Equity Market is measured by the mean rank across three variables used in La Porta et al. (1997): (1) the ratio of the aggregate stock market capitalization held by minorities to gross national product, (2) the number of listed domestic firms relative to the population, and (3) the number of IPOs relative to the population. Each variable is ranked such that higher scores indicate a greater importance of the stock market. Ownership Concentration is measured as the median percentage of common shares owned by the largest three shareholders in the ten largest privately owned non-financial firms (La Porta et al., 1998). The Disclosure Index measures the inclusion or omission of 90 items in the 1990 annual reports (La Porta et al., 1998).

	Outside Investor Rights	Legal Enforcement	Importance of Equity Market	Ownership Concentration	Disclosure Index
Aggregate Earnings Management	−0.538	−0.291	−0.418	0.434	−0.686
	(0.002)	(0.112)	(0.019)	(0.015)	(0.000)
Outside Investor Rights		−0.026	0.515	−0.344	0.568
		(0.888)	(0.003)	(0.058)	(0.002)
Legal Enforcement			0.522	−0.396	0.393
			(0.003)	(0.028)	(0.038)
Importance of Stock Market				−0.315	0.647
				(0.084)	(0.000)
Ownership Concentration					−0.398
					(0.036)

Table 2, Panel C: Correlations with institutional characteristics.

Table 3

Earnings management and institutional clusters

The table presents results from a k-means cluster analysis using three distinct clusters and nine institutional variables from La Porta et al., (1997, 1998). See Panel B of Table 2 for details. The variables are standardized to z-scores. Panel A reports the means of the institutional variables by cluster. Panel B reports the cluster membership for the 31 sample countries based on the cluster analysis performed on the variables in panel A. Countries in each cluster are sorted by the aggregate earnings management score from Panel A in Table 2. CD (CM) indicates a code-law (common-law) tradition. This variable is *not* used in the cluster analysis. Panel C reports the mean aggregate earnings management score for each cluster. The last row reports one-sided p-values for differences in the means of the aggregate earnings management across clusters using a t-test.

Panel A: Mean values of institutional characteristics by cluster

Institutional Variables	Cluster 1	Cluster 2	Cluster 3
Stock Market Capitalization	0.82	0.46	0.21
Listed Firms	49.56	18.58	9.50
IPOs	4.04	0.55	0.37
Ownership Concentration	0.34	0.37	0.50
Anti-Director Rights	4.50	2.62	2.90
Disclosure Index	74.38	66.67	58.13
Efficiency of Judicial System	9.78	9.04	5.50
Rule of law	9.02	9.07	5.65
Corruption Index	8.80	9.09	5.13
	Outsider features	↔	Insider Features

Panel B: Cluster membership of countries

Institutional variables	Cluster 1	Cluster 2	Cluster 3
Countries Sorted by Aggregate Earnings	Singapore (CM)	Austria (CD)	Greece (CD)
Management Score	Hong Kong (CM)	Taiwan (CD)	Korea (CD)
	Malaysia (CM)	Switzerland (CD)	Portugal (CD)
	UK (CM)	Germany (CD)	Italy (CD)
	Norway (CD)	Japan (CD)	India (CM)
	Canada (CM)	Belgium (CD)	Spain (CD)
	Australia (CM)	Netherlands (CD)	Indonesia (CD)
	USA (CM)	Denmark (CD)	Thailand (CM)
		France (CD)	Pakistan (CM)
		Finland (CD)	Philippines (CD)
		Sweden (CD)	
		South Africa (CM)	
		Ireland (CM)	

Panel C: Pervasiveness of earnings management by cluster

	Cluster 1	Cluster 2	Cluster 3
Mean Aggregate Earnings Management Score	10.1	16.1	20.6
Tests of EM differences between clusters	C1 vs. C2	C2 vs. C3	C1 vs. C3
(p-values)	(0.044)	(0.059)	(0.003)

Table 3: Earnings management and institutional clusters.

Table 4

Earnings management, outside investor protection and private control benefits

The table presents coefficients and two-sided p-values (in parentheses) from rank regressions with the Aggregate Earnings Management Measure as the dependent variable, which is created by averaging the ranks of all four earnings management measures, EM1–EM4 (see Table 2). Outside Investor Rights are measured by the anti-director rights index from La Porta et al., (1998), which ranges from zero to five. Legal Enforcement is measured as the average score across three legal variables used in La Porta et al., (1998): (1) the efficiency of the judicial system, (2) an assessment of rule of law, and (3) the corruption index. All three variables range from zero to ten. Private Control Benefits are measured at the country level as the average block premium estimated by Dyck and Zingales (2002) based on transfers of controlling blocks of shares. The first column presents a simple rank regression. The second regression is estimated using two-stage least squares. Instrumental variables are the rank of the country's real per capita GDP averaged from 1980 to 1989, and three binary variables indicating an English, German, French, or Scandinavian legal origin based on the classification in La Porta et al., (1998). The third regression is also estimated using two-stage least squares. The instrumental variables are the Outsider Rights Index and the Legal Enforcement.

	Aggregate Earnings Management Measure	Aggregate Earnings Management Measure - 2SLS -	Aggregate Earnings Management Measure - 2SLS -
Constant	28.605	31.421	3.128
	(<0.001)	(<0.001)	(0.463)
Outside Investor Rights	−0.499	−0.641	—
	(<0.001)	(0.001)	
Legal Enforcement	−0.289	−0.322	—
	(0.025)	(0.025)	
Private Control Benefits	—	—	0.931
			(0.004)
Adjusted R^2	0.389	0.359	0.272
Number of Observations	31	31	26

Table 4: Investor protection and private control benefits.

Table 5
Earnings management and outside investor protection: Controlling for differences in the accounting rules and ownership concentration
The table presents coefficients and two-sided *p*-values (in parentheses) from rank regressions of the Aggregate Earnings Management Measure on Outside Investor Rights and Legal Enforcement controlling for other institutional factors. Outside Investor Rights is the anti-director rights index from La Porta et al., (1998), which ranges from zero to five. Legal Enforcement is measured as the average score across three legal variables used in La Porta et al. (1998): (1) the efficiency of the judicial system, (2) an assessment of rule of law, and (3) the corruption index. All three variables range from zero to ten. The Accrual Rules variable captures the extent to which accrual rules accelerate the recognition of economic transactions (e.g., R&D activities or pension obligations) in accounting. It is constructed by Hung (2001). Ownership concentration is measured as the median percentage of common shares owned by the largest three shareholders in the ten largest privately owned non-financial firms (La Porta et al., 1998). The regressions in columns 2 and 4 are estimated using two-stage least squares. Instrumental variables are the rank of the country's real per capita GDP averaged from 1980 to 1989, and three binary variables indicating an English, German, French, or Scandinavian legal origin based on the classification in La Porta et al. (1998).

	Aggregate Earnings Management *Controlling for Accounting Rules*	Aggregate Earnings Management *Controlling for Accounting Rules* - 2SLS -	Aggregate Earnings Management *Controlling for Ownership*	Aggregate Earnings Management *Controlling for Ownership* - 2SLS -
Constant	30.974	34.591	24.333	47.261
	(<0.001)	(<0.001)	(<0.001)	(0.002)
Outside Investor Rights	−0.285	−0.501	−0.444	−0.774
	(0.079)	(0.044)	(0.003)	(0.007)
Legal Enforcement	−0.297	−0.420	−0.228	−0.571
	(0.080)	(0.048)	(0.101)	(0.048)
Accrual Rules	−0.689	−0.425	—	—
	(0.016)	(0.313)		
Ownership Concentration	—	—	0.151	−0.609
			(0.302)	(0.225)
Adjusted R^2	0.584	0.468	0.392	0.214
Number of Observations	20	20	31	31

Table 5: Accounting rules, ownership concentration, and robustness.

9 Short summary

- The paper studies systematic differences in earnings management across 31 countries.
- The main hypothesis is that insiders use earnings management to conceal private control benefits, so earnings management should be lower where outside investors are better protected.
- The sample contains 70,955 firm-year observations for 8,616 nonfinancial firms from 1990 to 1999.
- Earnings management is measured using smoothing, accrual magnitude, and small-loss avoidance proxies.
- The aggregate earnings-management score is highest in countries such as Austria, Greece, South Korea, Portugal, and Italy.
- The score is lowest in countries such as the United States, Australia, Ireland, Canada, Norway, Sweden, and the United Kingdom.
- Cluster analysis shows that outsider economies with strong markets and enforcement have the lowest earnings management.
- Rank regressions show that outside investor rights and legal enforcement are negatively related to earnings management.

- Private control benefits are positively related to earnings management.
- The results are robust to controls for economic development, firm characteristics, accounting rules, and ownership concentration.
- The paper's key contribution is to make earnings quality endogenous to investor protection and corporate governance.

10 Conclusion

10.1 Core conclusion

Leuz, Nanda, and Wysocki show that earnings management differs systematically across countries and is lower where outside investors are better protected. Countries with stronger shareholder rights, better legal enforcement, large equity markets, and more dispersed ownership have more credible reported earnings.

10.2 Economic intuition

Insiders manage earnings because opaque reporting helps them conceal private control benefits. Strong investor protection reduces the amount insiders can divert and improves outsiders' ability to discipline insiders. As a result, insiders have less reason to hide performance through smoothing, accrual manipulation, or loss avoidance.

10.3 Incremental contribution revisited

The paper extends the law-and-finance literature by showing that legal institutions affect not only corporate financing, dividend policy, and ownership structure, but also the quality of accounting numbers. Financial reporting quality is not just a technical accounting feature; it is part of the governance equilibrium.

10.4 Implications

The findings imply that improving accounting standards alone may be insufficient if enforcement and investor rights remain weak. Credible reporting depends on the institutional environment that makes poor reporting costly and makes outside discipline feasible.

10.5 Limitations

The study is cross-country and observational, with a small number of countries. Earnings management is difficult to measure, and the proxies may capture different reporting practices across institutional settings. Legal institutions, disclosure, ownership, and financial development are complementary and hard to disentangle fully.

10.6 Takeaway

The enduring takeaway is that investor protection shapes earnings quality. Strong legal rights and enforcement reduce insiders' incentives to conceal firm performance, while weak protection allows accounting opacity to persist as a tool for preserving private control benefits.